AN8280G-TK Amplifier Replacement Procedure

AMPLIFIER REPLACEMENT

Step 1: Preparation

- Turn AC power OFF at service panel.
- Remove front and rear doors.
- Turn power OFF at PSU at rear of cabinet.
- Verify all LEDs are OFF on front of PSU.
- Wait 5 minutes.

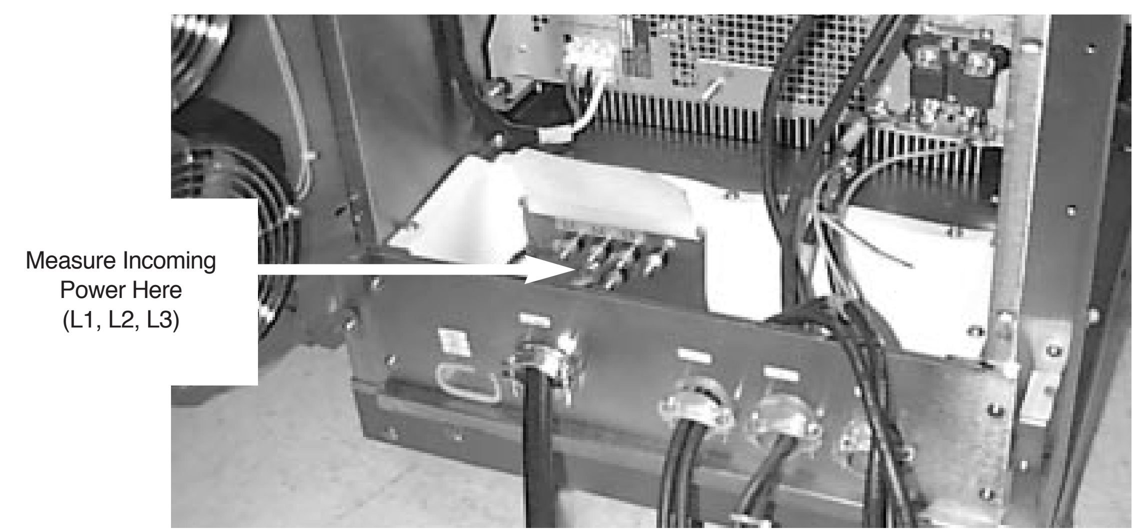
LOCKOUT/TAGOUT PROCEDURE

This material is to be applied to all replacement procedures that involve the Scaleable Gradient Cabinet modules.

DANGER!!

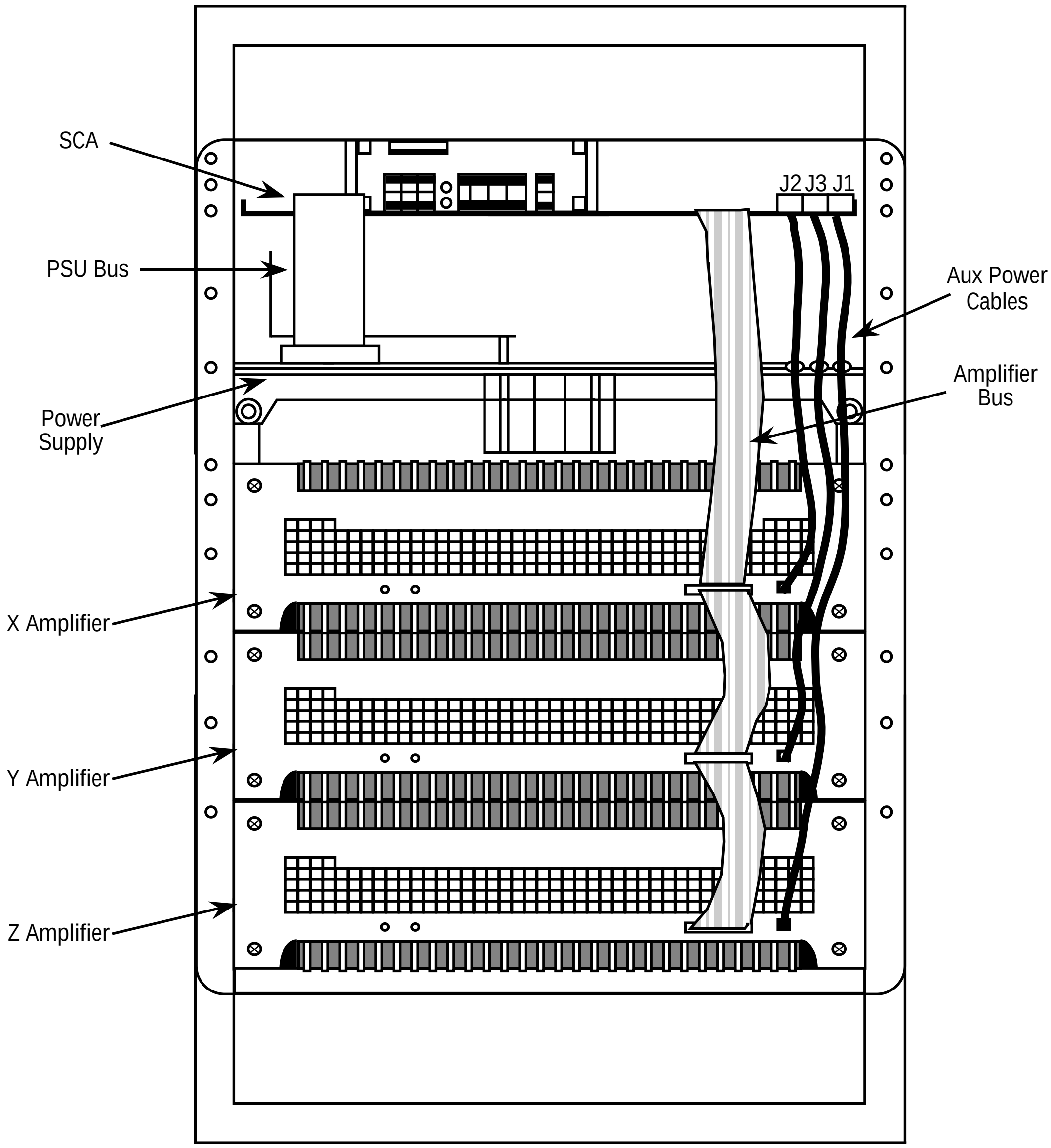
FATAL ELECTRIC SHOCK HAZARD! TO PREVENT FATAL ELECTRIC SHOCK, DISCONNECT POWER FROM THE PDU BEFORE YOU PERFORM THE FOLLOWING PROCEDURES. PERFORM LOCKOUT/TAGOUT PROCEDURE PER GE OSHA LOCKOUT/TAGOUT REQUIREMENTS 29 CFR 1910.147. DO THIS BY SECURING THE PDU CIRCUIT BREAKER FOR THE SCALEABLE GRADIENT CABINET.

1. Turn off the SGD Cabinet Breaker at the Power Distribution Unit. Lock out the Breaker and tag it.
2. After power to the SGD Cabinet has had sufficient time to dissipate, take a Digital Multimeter and set it to its highest AC voltage range.
3. Verify that all energy has been dissipated by measuring incoming power to all components of the Scaleable Gradient Cabinet.
   - Place the reference probe (black) on the SGD Cabinet Ground.
   - Locate L1, L2, and L3. These are the 208V, 3 Phase inputs to the SGD Base SR-20 Cabinet.
4. Measure voltage at each of three 208 volt input terminals. The meter should read 0 volts AC at each of the three measuring points. See below.

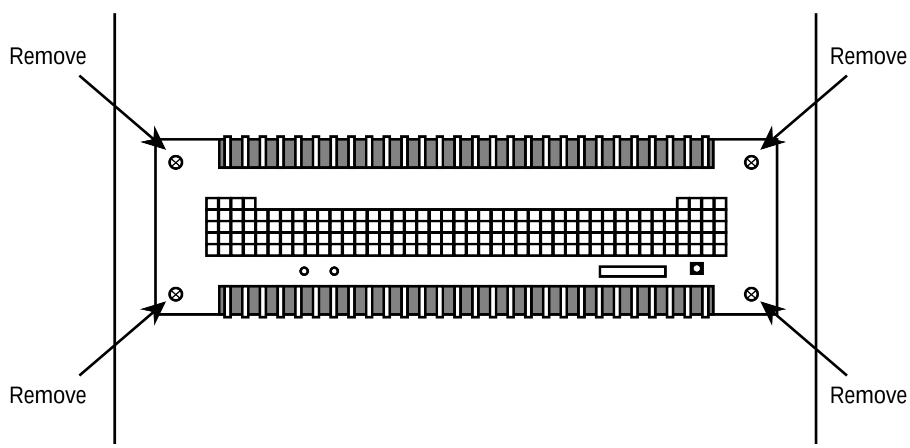


Step 2: Disconnect Amp Bus and Auxiliary Power Cables at Front of Amplifier to be Replaced.

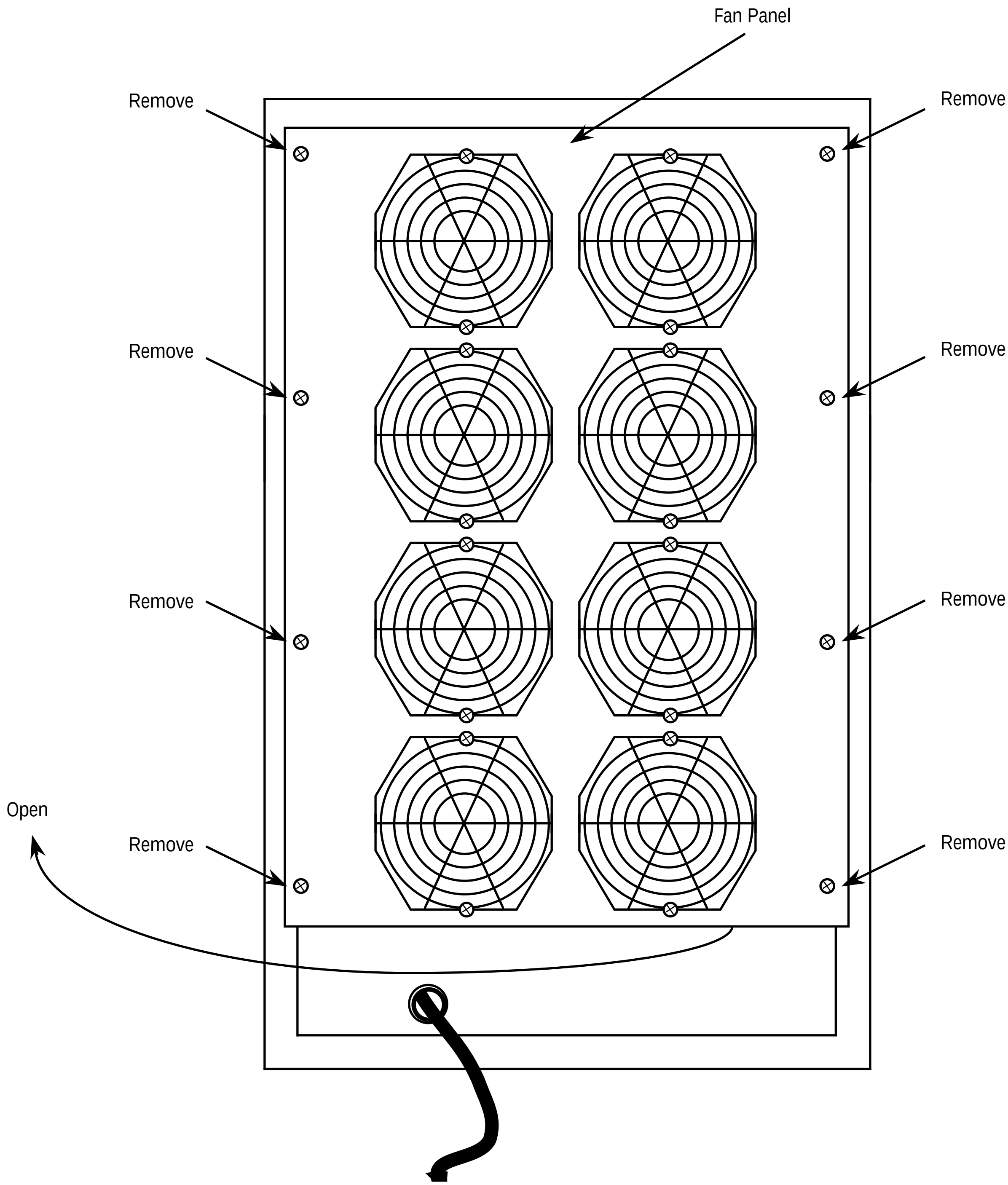
• Disconnect all other control ribbon cables to amplifiers and drape over top of rack.



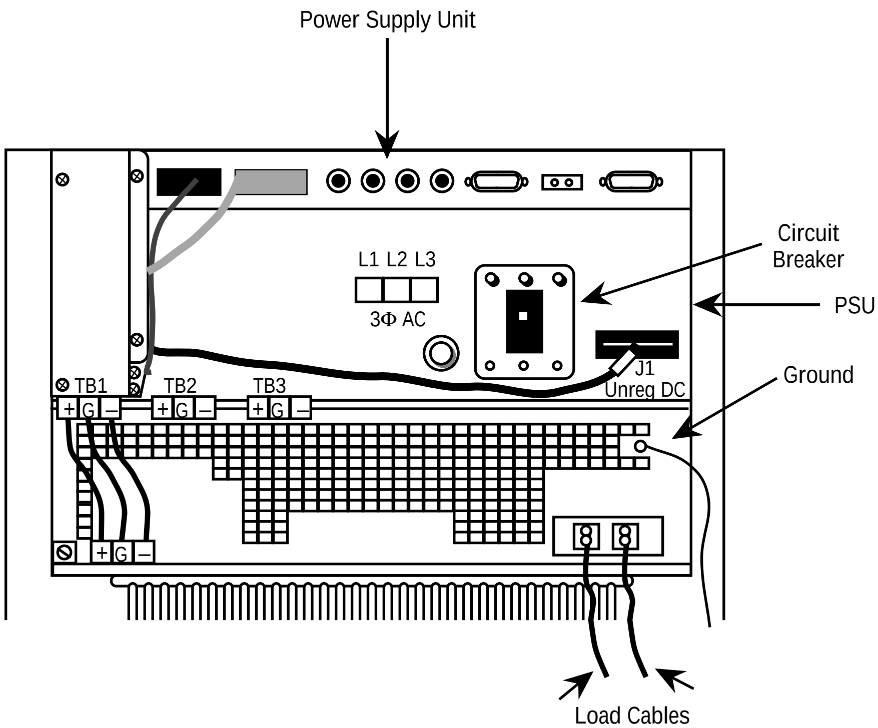
Step 3: Remove 4 Screws Holding Front Panel of Amplifier to Rack.



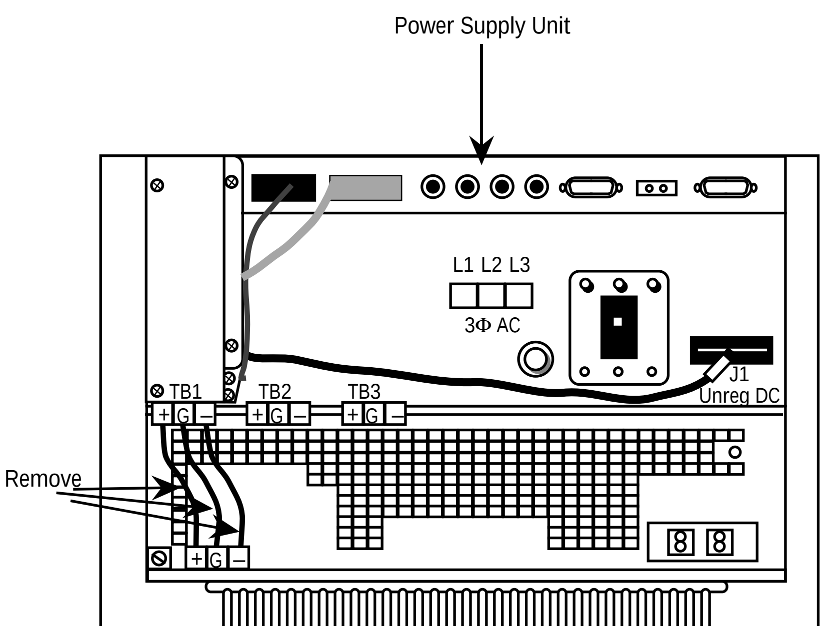
Step 4: At Rear of Rack, Remove 8 Screws and Open Fan Panel.



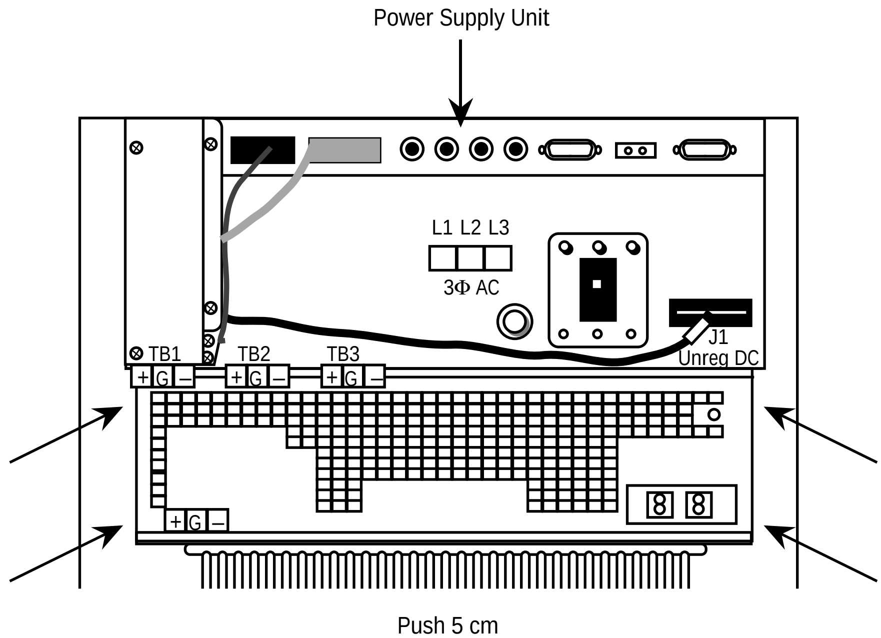
Step 5: Remove 2 GEYMS Load Cables from Amplifier, Including 1 Load Ground.



Step 6: Remove 3 H.V. DC Power Cables from Amplifier to be Replaced. Use the Special Screwdriver Provided.



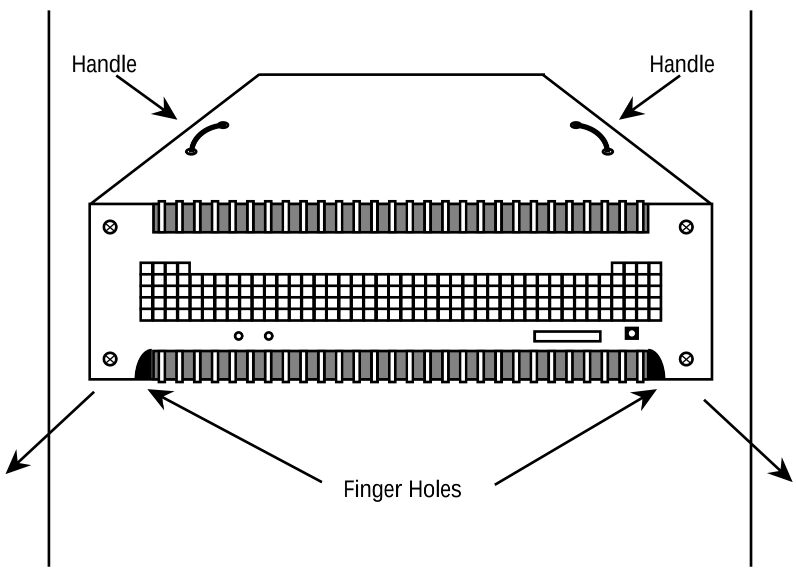
Step 7: Push Amplifier Forward in Rack 5 cm by Pushing on Rear of Amplifier.



Step 8: Pull Front of Amplifier out of Front until Handles are Exposed.

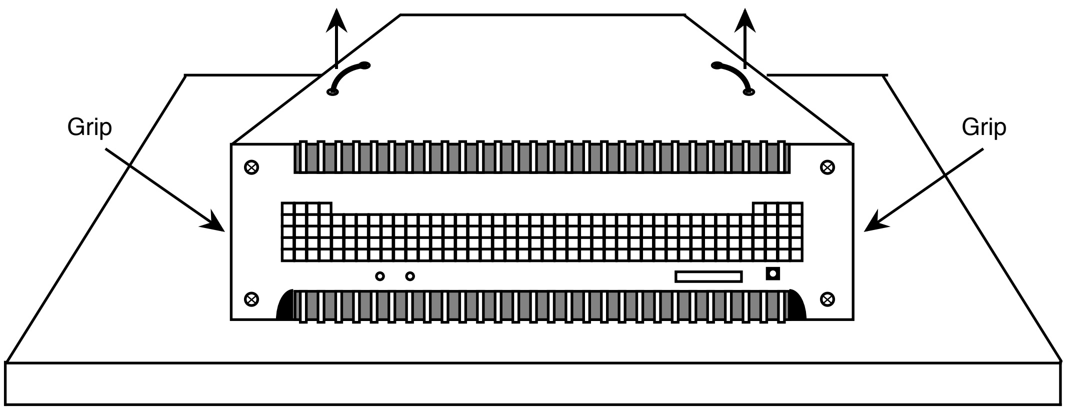
- Insert handles if needed.

CAUTION: Amplifier weight 28 kg. Must be supported if pulled out past center of gravity.



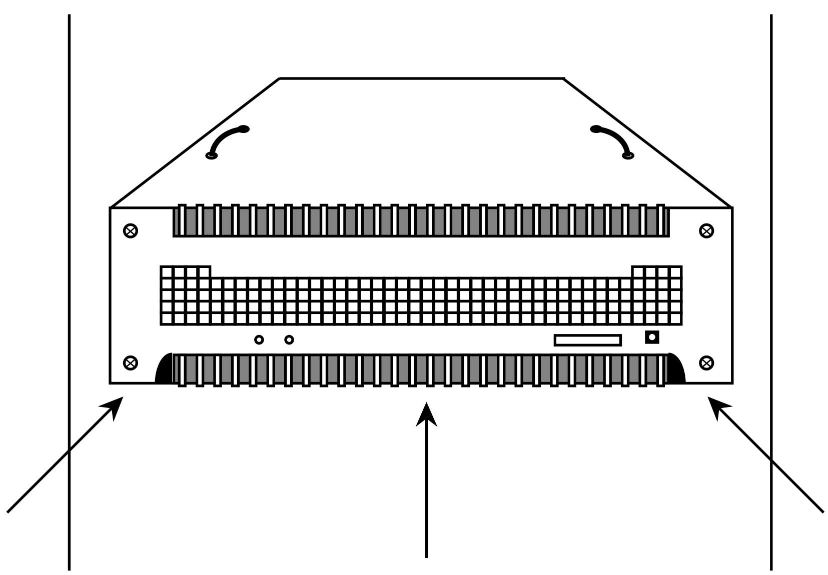
Step 9: Remove Amplifier from Front.

- Carry by handles or sides. Be careful to avoid damage to terminals.
- Lay amplifier on bottom heat sink. Never place amplifier on front or rear, as connections may be damaged.



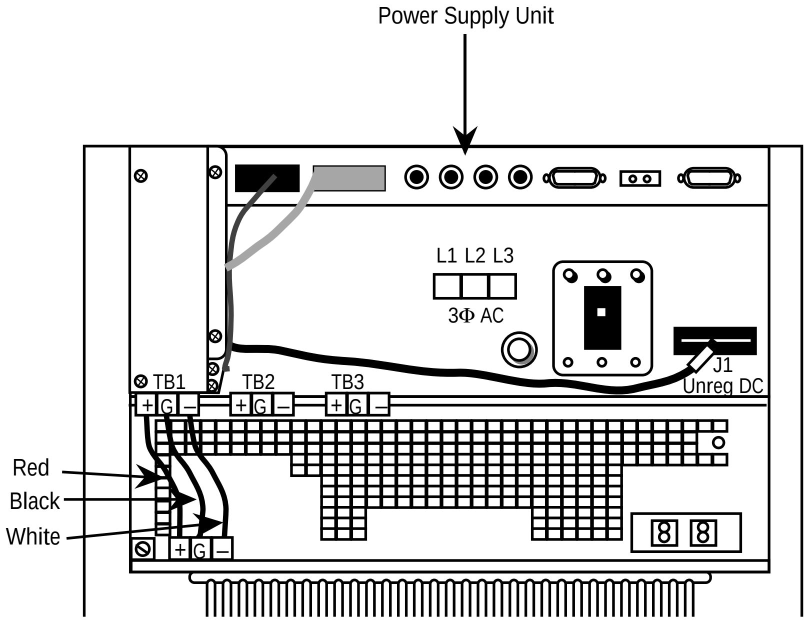
Step 10: Place New Amplifier in Rack and Slide in from Front to Back.

- Be sure front screws line up with holes.
- Engage guide pins.



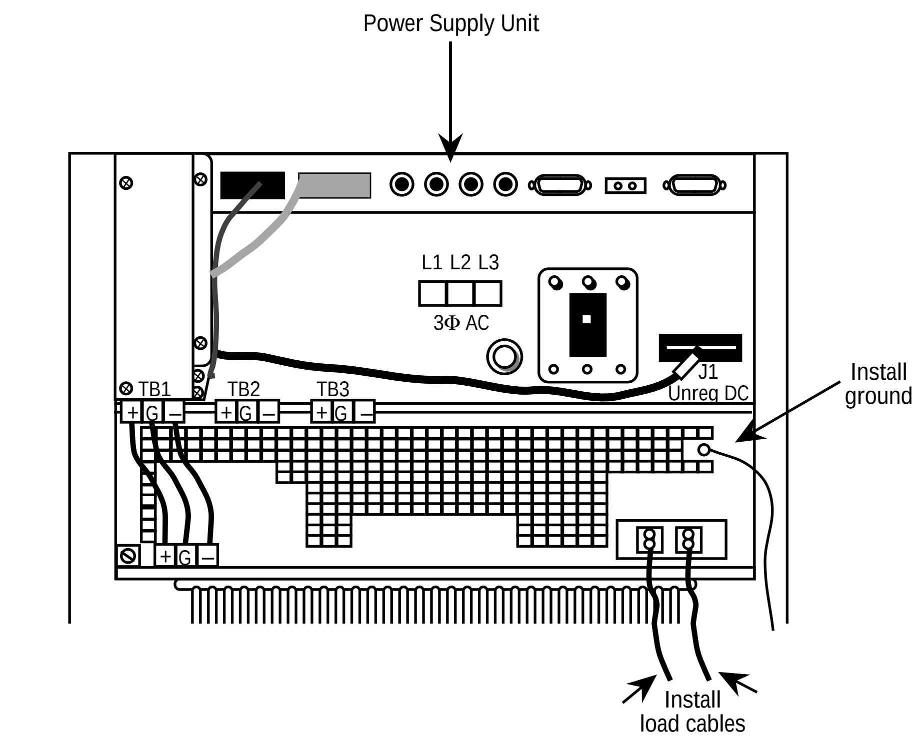
Step 11: Install 3 H.V. Cables in Amplifier Terminals.

• Tighten with special screwdriver provided. Do not damage wire connections.

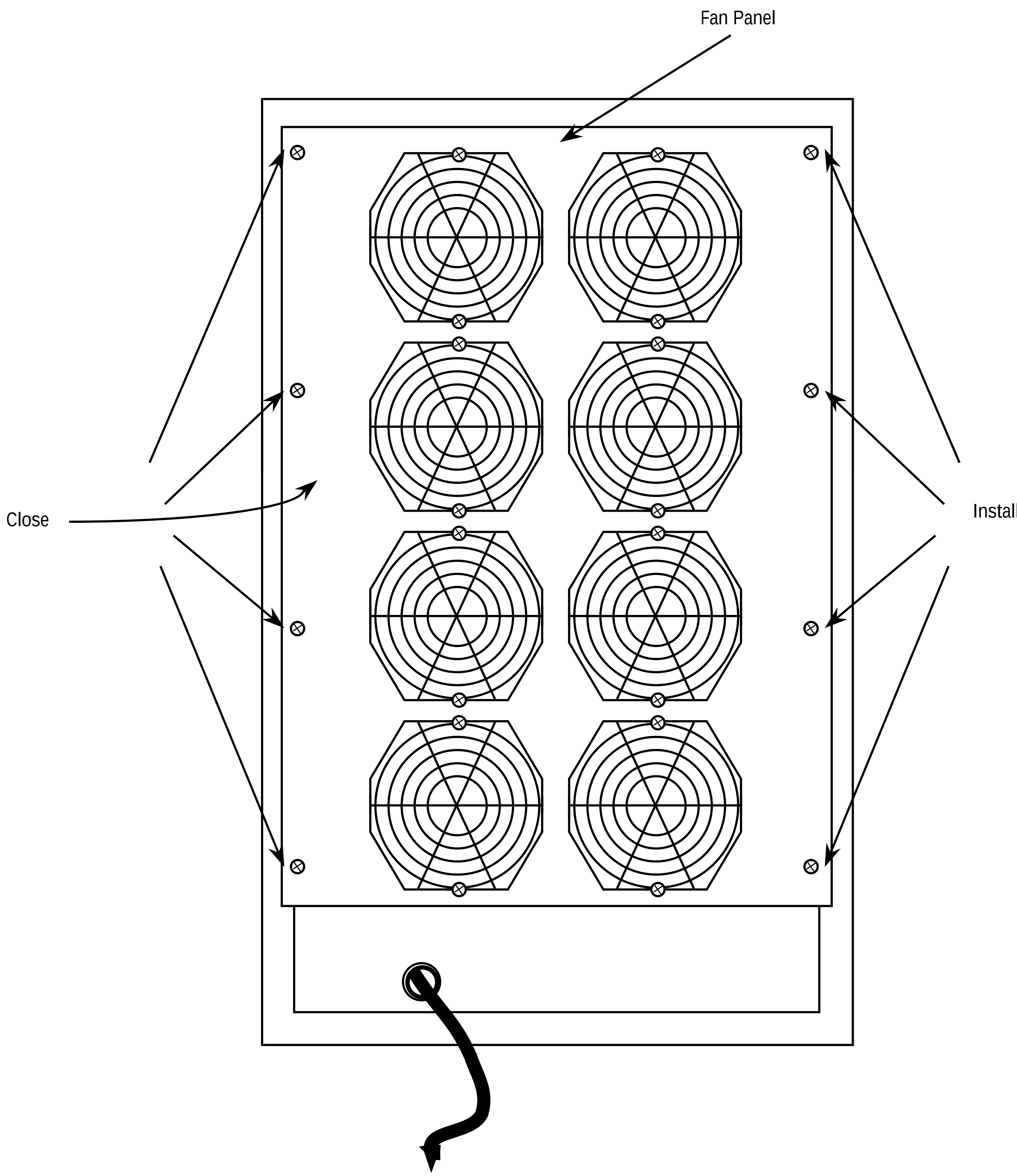


Step 12: Install 2 GEYMS Load Cables plus Ground Load and Ground Terminals.

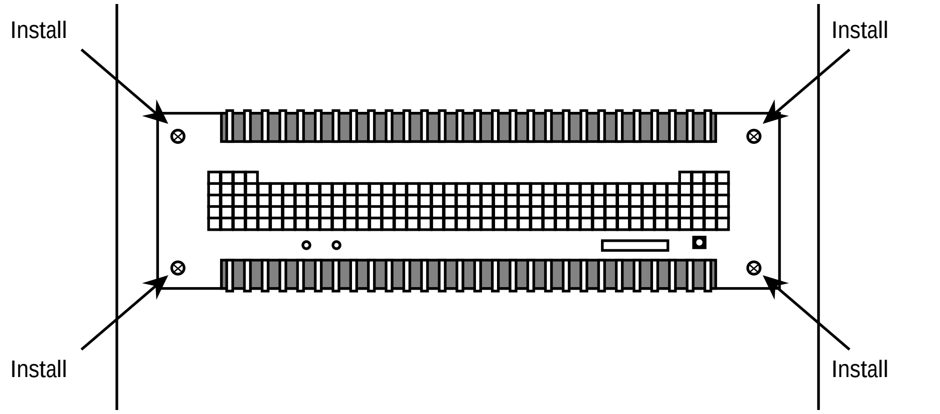
• Inspect and verify all connections are visually correct. Observe polarity of cables.



Step 13: Close Fan Panel and Install 8 Screws.

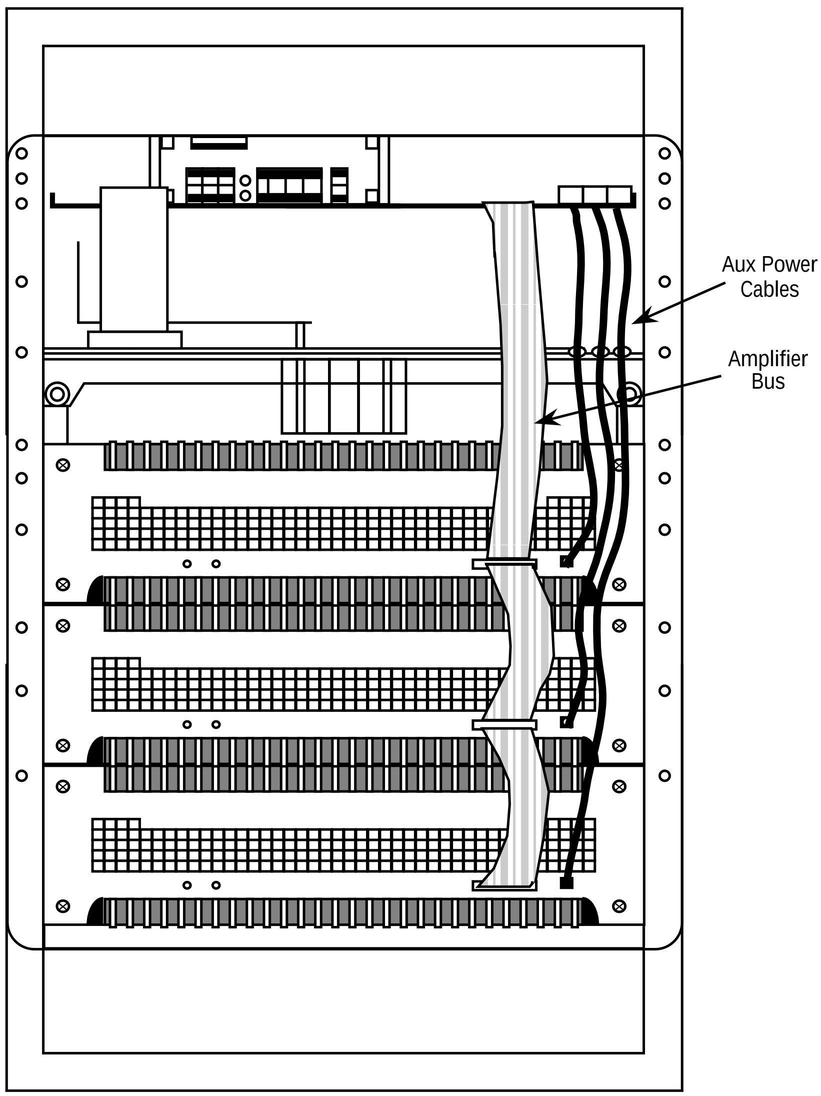


Step 14: Install 4 Front Panel Screws Holding Amplifier to Rack.



Step 15: Install Amplifier Bus and Auxiliary Power Cables on Front of Amplifier.

- Reconnect control ribbon cable to all amplifiers. Be sure connectors are latched.



Step 16: Inspect Unit Prior to Turn-On.

- Turn on AC circuit breaker on rear of PSU. Turn on AC power at service panel. Verify proper power-up sequence. Replace front and rear doors.

FUNCTIONAL CHECKS REQUIRED

1. Check grafidy using procedure for Eddy Current Compensation for Signa Horizon on the axis with the replacement amplifier and calibrate if out of specification.

2. Check gradient calibration using procedure for Gradient Calibration (DQA Version) or alternate proprietary procedure for SPT Quick Head Check; calibrate if necessary.

3. Check shim using procedure for LVShim Check; shim if necessary.